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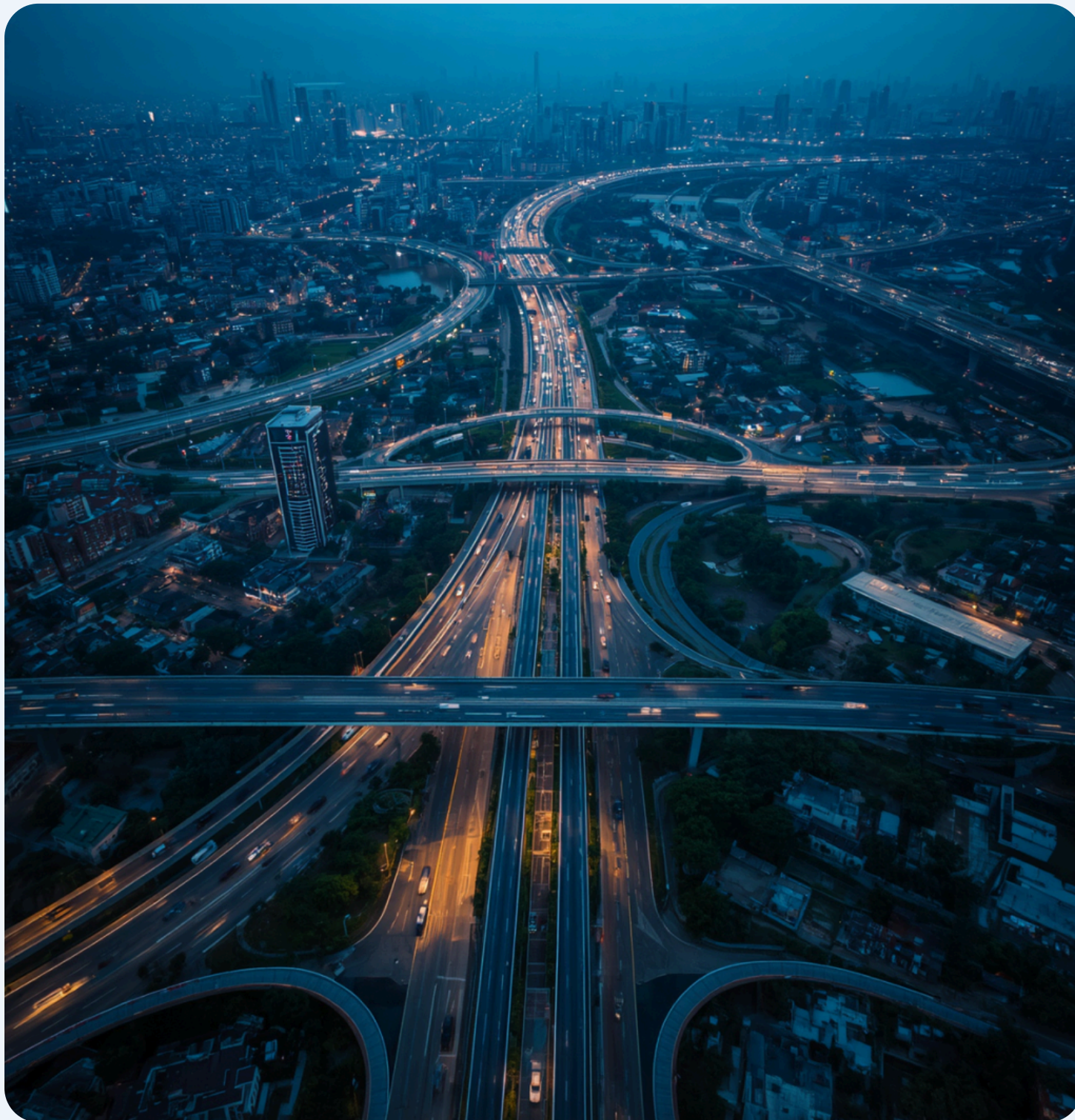
Urban Pulse – Smart City Mobility Intelligence Platform



Dataset Collection & Data Source Strategy

Tagline:

A unified approach to collecting,
integrating, and analyzing smart city
mobility data.



Why Bengaluru?

Bengaluru was selected for its rich, publicly available smart city datasets, making it ideal for a comprehensive mobility intelligence platform.

📍 Key Data Sources:

- BMTC GTFS – Bus routes, stops & timetables
- Traffic Data – Congestion levels & speed
- Weather Data – Temperature, rainfall & wind
- Road Accident Data – Location & severity
- Census Data – Population & literacy metrics
- BBMP GeoJSON – Ward boundaries & maps
- OpenStreetMap GIS – Roads, buildings & routes

✅ Outcome: Enables a 360° transportation, infrastructure, and mobility analytics ecosystem.

Dataset Overview



Weather Dataset

Contains: Temperature, Rainfall, Wind Speed
→ Used For: Weather Impact on Traffic & Transit



Traffic Dataset

Contains: Congestion Level, Vehicle Count, Speed
→ Used For: Traffic Trends & Congestion Analysis



Census Dataset

Contains: Population, Literacy, Workforce →
Used For: Mobility Demand & Equity Analysis



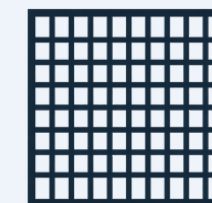
BMTG GTFS Dataset

Contains: Bus Routes, Stops, Trips, Timetables
→ Used For: Transit Analytics & Schedule Analysis



Accident Dataset

Contains: Location, Severity, Date & Time →
Used For: Hotspot Detection & Road Safety Analysis



BBMP & OpenStreetMap GIS

Contains: Ward Boundaries, Roads, Buildings
→ Used For: Maps, Heatmaps & Route Optimization

How We Use These Datasets

01

Raw Datasets → Data Cleaning
Standardize formats, remove duplicates, handle missing values

02

Power BI Dashboard
Build interactive dashboards for mobility and transit analytics

03

Data Integration → Central Database
Merge datasets by location and time into unified warehouse

04

AI Analytics & Forecasting
Apply predictive models for traffic, transit, and urban planning

Data Pipeline

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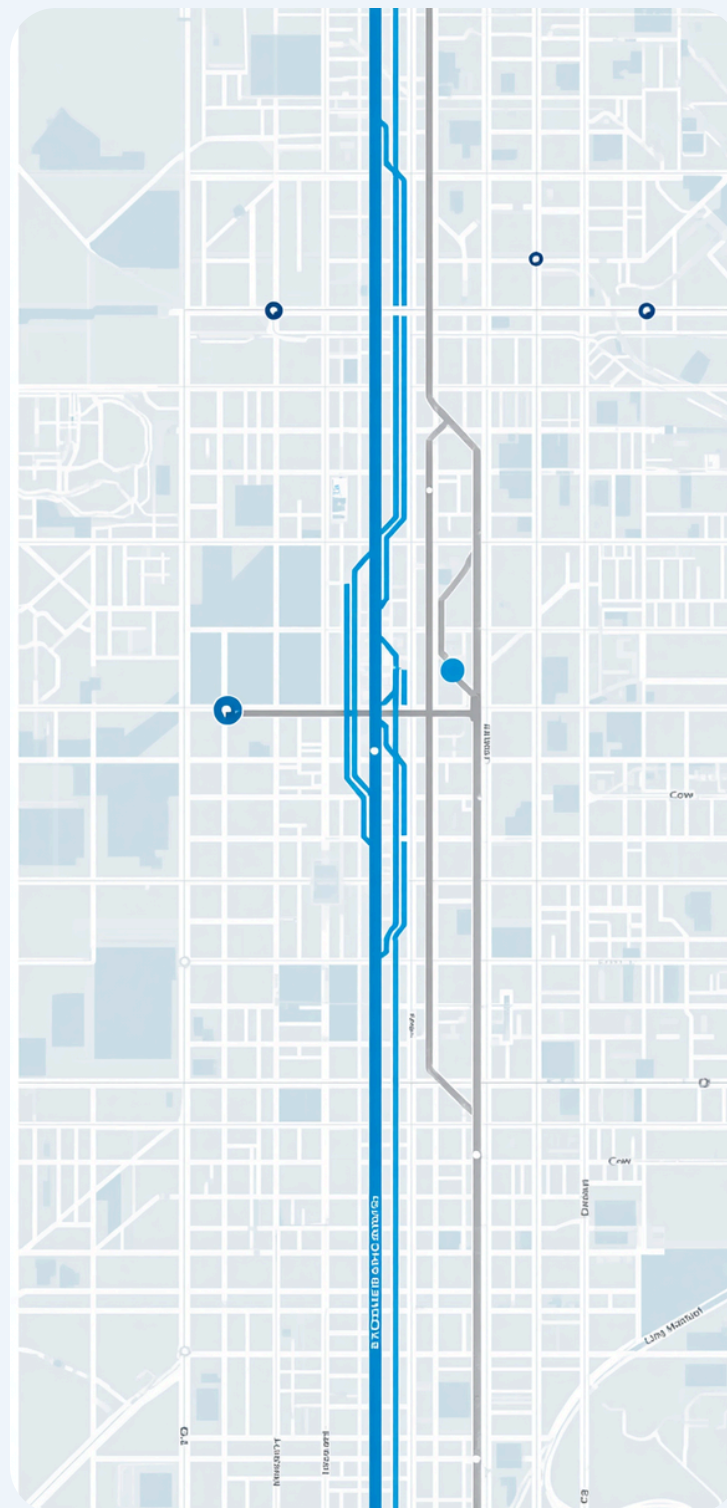


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BMTC GTFS Dataset



Bus Routes & Stops

Detailed bus route paths, stop locations, trip sequences, timetables, and fare information for Bengaluru's public transit network.



Schedule Analysis

Analyze trip schedules, frequency patterns, and timetable adherence to optimize transit service planning.



Transit Performance

Monitor on-time performance, ridership trends, and service efficiency across BMTC bus corridors.



Accessibility Analysis

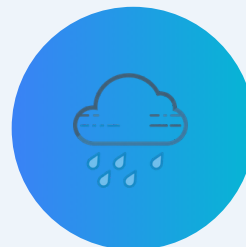
Evaluate transit coverage, proximity to stops, and equitable service distribution across city wards.

Traffic, Weather & Accident Datasets



Traffic Dataset

Congestion Level, Vehicle Count, Peak Hours, Average Speed → Traffic trends & congestion analysis.



Weather Dataset

Temperature, Rainfall, Wind Speed → Weather impact on traffic and transit operations.



Accident Dataset

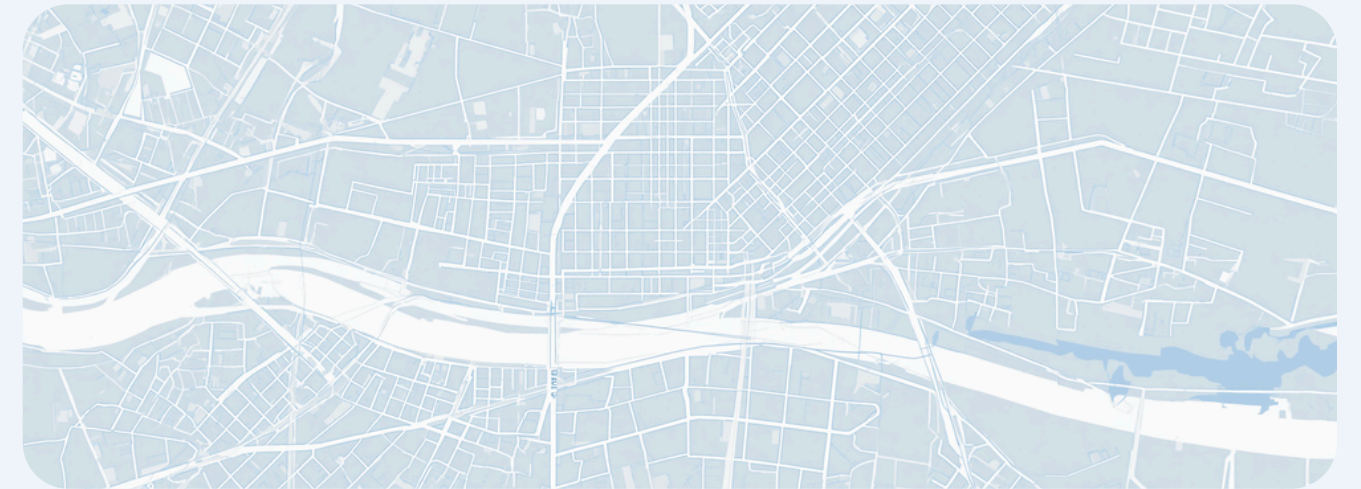
Accident Location, Severity, Date & Time → Hotspot detection & road safety analysis.



Integrated Analytics

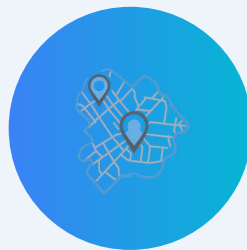
Combining traffic, weather, and accident data enables predictive urban mobility intelligence.

Census, BBMP & GIS



Census Dataset

Population, Households, Literacy, Workforce data enabling mobility demand & equity analysis across Bengaluru.



BBMP GeoJSON

Ward boundary geometries used for ward-wise heatmaps, spatial aggregation, and geographic mapping.



OpenStreetMap GIS

Roads, buildings, railways, and bus stops enabling route optimization and geospatial analysis.



Integrated Geospatial View

Combining demographic, boundary, and infrastructure layers for a 360° urban mobility perspective.

Live API Integration

Live APIs complement historical datasets by providing real-time city conditions. A dedicated Live City Status Dashboard displays current data without impacting historical analysis or forecasting pipelines.



Historical Datasets

Used for in-depth analytics and forecasting. Historical records from 2025 form the backbone of predictive models.

Weather API

Real-time weather conditions including temperature, rainfall, and wind speed streamed live into the dashboard.

Traffic API

Live congestion levels, vehicle counts, and average speed data to monitor current road conditions across the city.

Transit API

Real-time bus routes, stop arrivals, and trip status to power the Live City Status Dashboard separately.

Complete Architecture

Historical Data Flow

Historical Datasets → Data Cleaning → Data Integration → Data Warehouse

Data Processing Layer

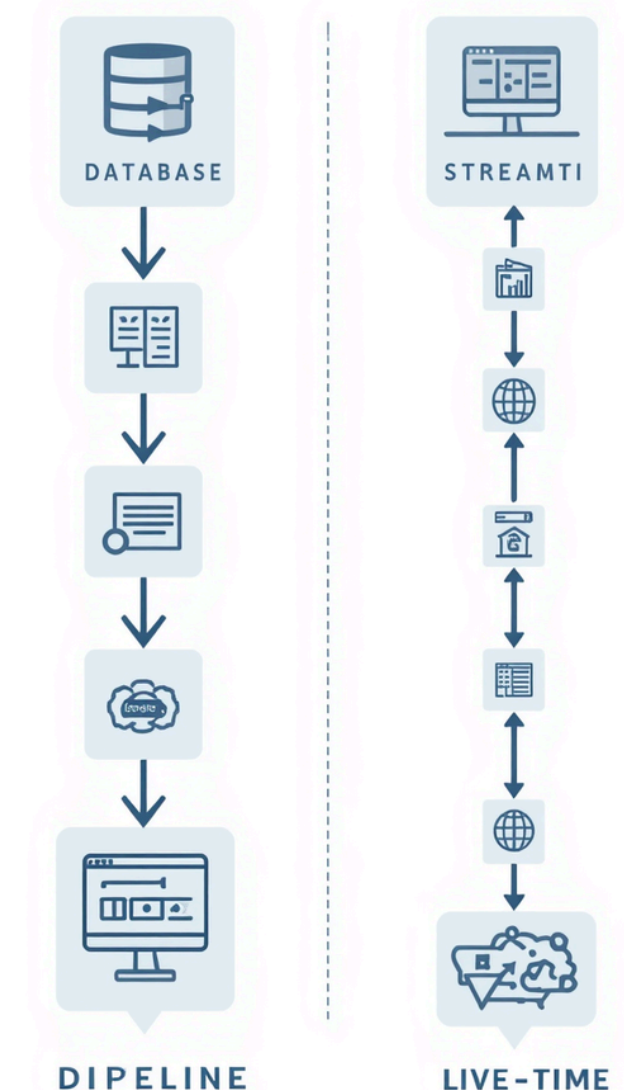
Standardize, merge by location & time, build central repository for reliable analytics.

Live API Flow

Live APIs (Weather, Traffic, Transit) → Backend → Live City Status Dashboard

Output & Analytics

Power BI Dashboard → AI Analytics → Forecasting & Smarter Urban Planning



Conclusion

- Bengaluru offers comprehensive, publicly available smart city datasets for robust urban analytics.
 - Historical (2025) datasets provide reliable, structured data for trend analysis and forecasting.
 - Live APIs enhance the platform with real-time traffic, weather, and transit monitoring.
 - Integrated transportation, weather, demographic, and GIS data create a 360° urban mobility view.
- "Urban Pulse combines historical analytics with real-time intelligence to support smarter, data-driven urban mobility planning."

